

# 9.1: Binomial Expansions;

Binomial Coefficients form the pattern of pascals Triangle.

Binomial: $(a+b)^n$, where $n$ can be seen as a row in Pascal's Triangle.

Going along the row, you increase a value named '$r$'. $n=2$, $r=0$, $r=1$ for $r=0$, $r=1$. There are $n+1$ values of $r$ for row $n$, from $0 \to n$

$n=5$: 1 5 10 10 5 1 $\Rightarrow \binom{n}{r} = \binom{n}{n-r}$. Value can be shown from form ${}^nC_r$ or $\binom{n}{r}$.

$r=0, 1, r=2, 1=3, r=4, r=5$.

${}_nC_0 = 1$, ${}_nC_1 = \frac{n}{1}$, ${}_nC_2 = \frac{n(n-1)}{2\times 1}$, ${}_nC_3 = \frac{n(n-1)(n-2)}{3\times 2\times 1}$ $\Rightarrow {}^nC_r = \frac{n!}{r!(n-r)!}$ / $\frac{n(n-1)(n-2)\dots(n-r+1)}{r!}$

$\Rightarrow {}^nC_{r+1} = \frac{n!}{r+1!(n-r-1)!}$ ${}_nC_r = \frac{({}^nC_{r-1})(n-r+1)}{r}$

last value $\times$ (row − (desired column) + 1) / desired column $\Rightarrow {}^nC_r = \frac{n!}{r!(n-r)!}$

$\Rightarrow (1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 \dots + \binom{n}{n-3}x^{n-3} + \binom{n}{n-2}x^{n-2} + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n$

$\binom{n}{3}$ $\binom{n}{2}$ $\binom{n}{1}$ $\binom{n}{0} = 1$

$\Rightarrow (1+x)^n = \sum_{r=0}^{n}\binom{n}{r}x^r = \sum_{r=0}^{n}\frac{n!}{r!(n-r)!}x^r$

$\Rightarrow (a+b)^n = \sum_{r=0}^{n}\binom{n}{r}a^n b^{n-r} = \sum_{r=0}^{n}\frac{n!}{r!(n-r)!}a^n b^{n-r}$ for $n \in \mathbb{Z}^+$

KEY:

$(1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 \dots + x^n$

$\binom{n}{r}a^r b^{n-r}$

$\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Approximations $(1+x)^n$, where $|x|<1$ & $n \in \mathbb{Q}$, truncated to the first few terms.

Relationships between Binomial Coefficients:

${}^nC_r = {}^nC_{n-r}$

${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$

When $n \notin \mathbb{N}$, expansion is infinite and only valid for $|x|<1$.

When $n \notin \mathbb{N}$, $(a+x)^n \to a^n\left(1+\frac{x}{a}\right)^n$

# 9.2: Selections and Permutations

Combinations → Order does not matter ($AB = BA$). ${}^nC_r = \frac{n!}{r!(n-r)!}$. $n$ = total things. $r$ = amount picked

Permutations → Order (of elements) does matter ($AB \neq BA$). ${}^nP_r = \frac{n!}{(n-r)!}$

Imagine picking 5 things out of 8. $\Rightarrow$ ($8\times7\times6\times5\times4$) ways of picking

$8! = 8\times7\times6\times5\times4\times3\times2\times1$

$(8-5)! = 3! = 3\times2\times1$

$\Rightarrow \frac{8!}{(8-5)!}$ = ways. $\Rightarrow \frac{n!}{(n-r)!}$. Permutation Formula.

$r!$ is your way to reorder your groups of things. If order does not matter, they are all the same $\Rightarrow {}^nC_r = \frac{{}^nP_r}{r!}$

$\Rightarrow {}^nC_r = \frac{n!}{r!(n-r)!}$